

Lander Minimum Time

P.24 Robot Motion, Simulation, Optimization

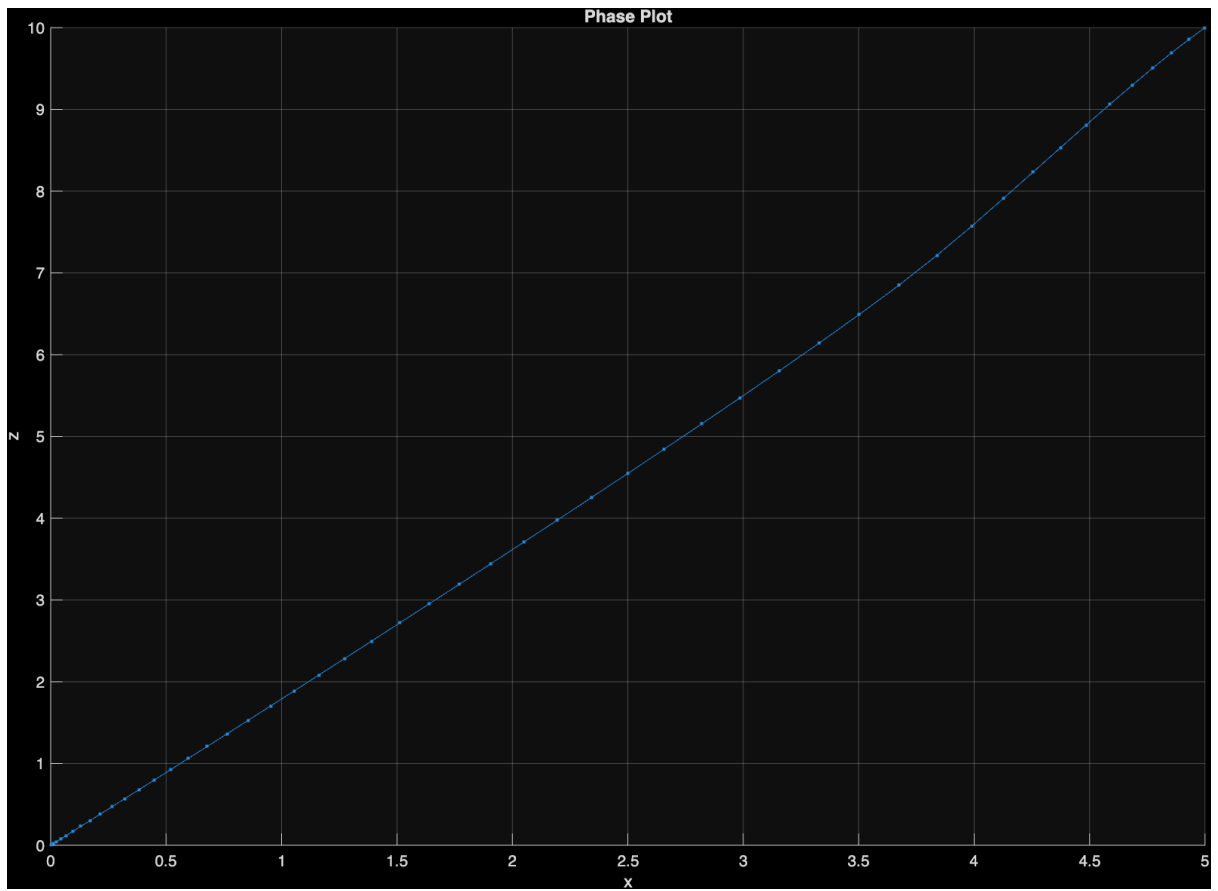
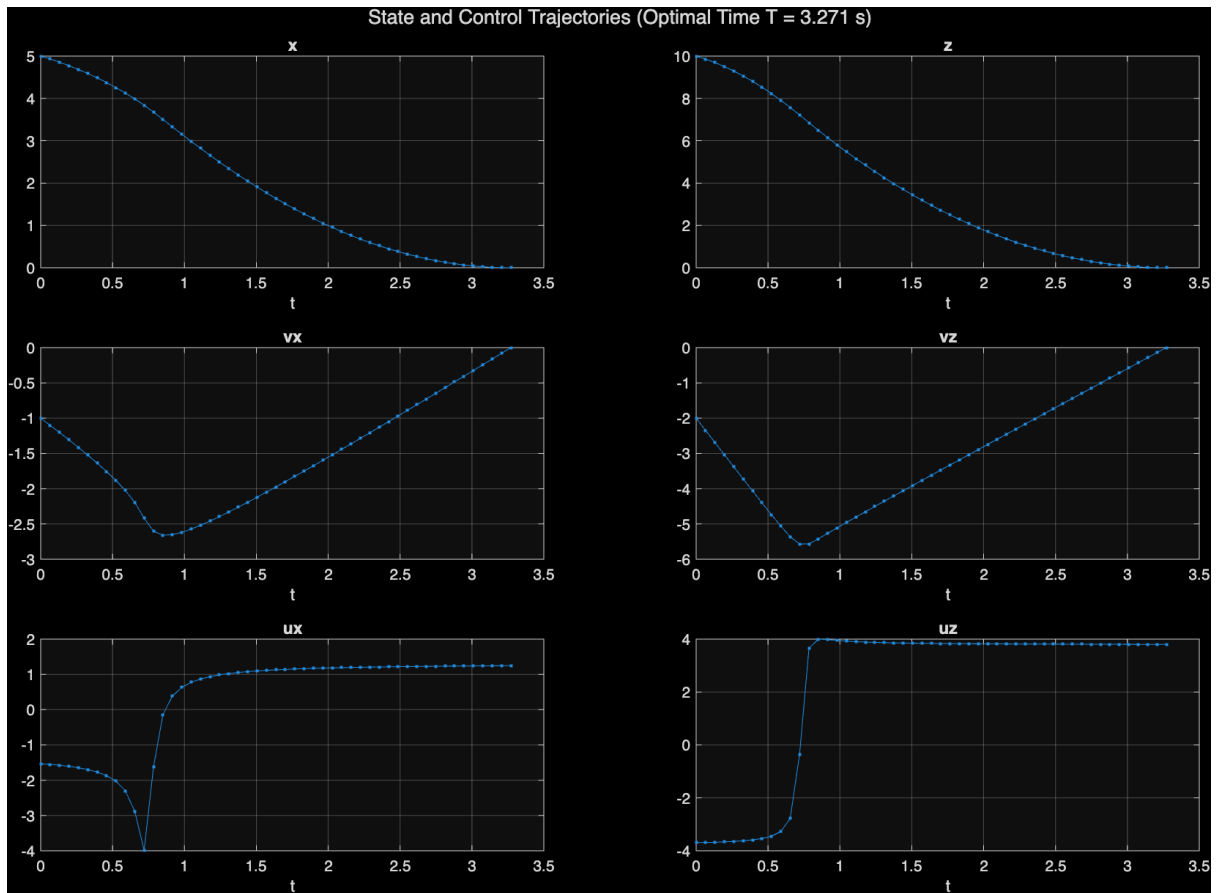
Suraj Dayma May 19, 2026

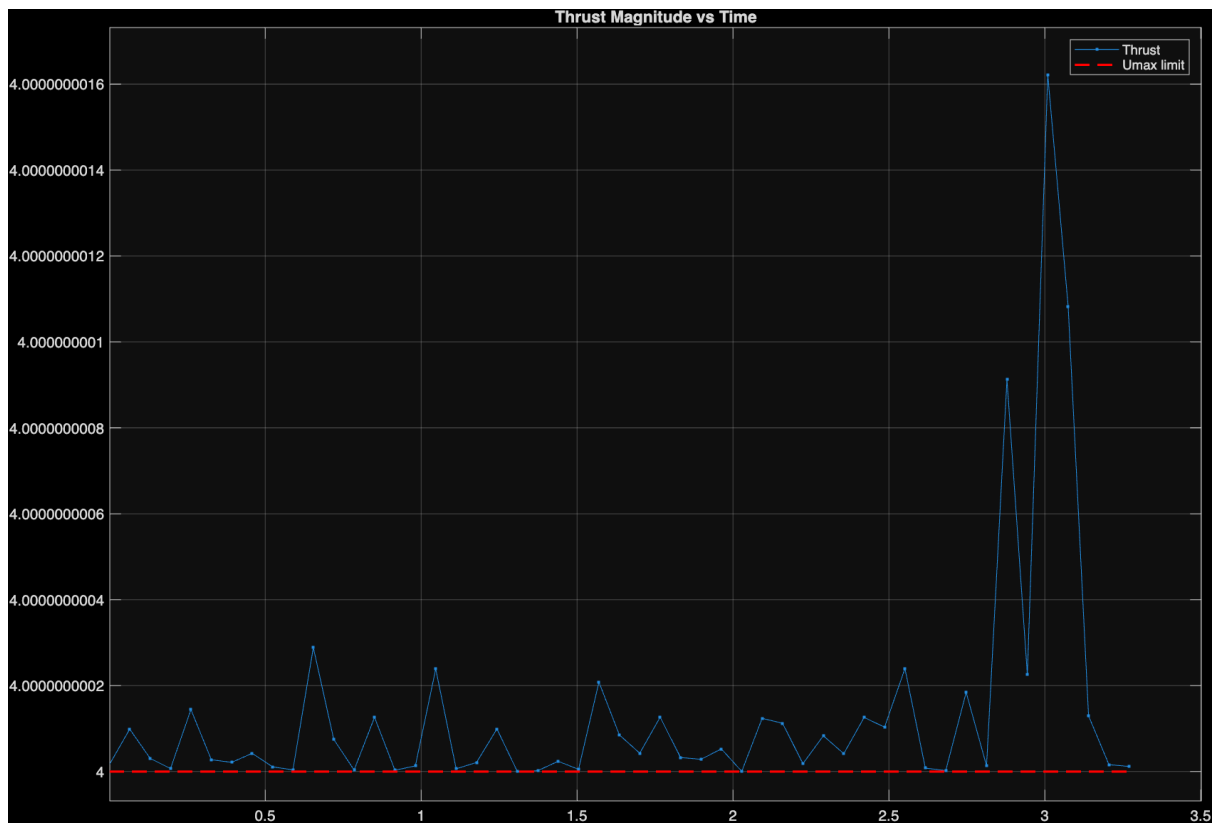
1. Problem

- SDolve the 2d point mass lander problem as a minimum time problem.
- I added the time T as a parameter for `fmincon` to find.
- A lower bound for time of $T \geq 0.1$ was also set.
- The number of points, N was fixed at 50.

2. Results

For $u_{\max} = 4$, $p.R = \text{diag}([1, 1])$; $p.Q = \text{diag}([1, 1, 1, 1])$; and same initial conditions as last problem.





- The trust magnitude is at umax the whole time within the numerical error.